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DOORBELL COMMUNITIES

Abstract

Doorbells can send data to each other to enable a first doorbell user to warn a second doorbell user regarding a suspicious visitor. A first user can indicate a first trait of a visitor via a first remote computing device. The first user can create a user group to enable the members of the user group to use their doorbells to take pictures of suspicious visitors and to send the pictures of the suspicious visitors to other members of the user group.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The entire contents of the following application are incorporated by reference herein: U.S. Nonprovisional patent application Ser. No. 18/132,072; filed Apr. 7, 2023; and entitled DOORBELL COMMUNITIES. [0002] The entire contents of the following application are incorporated by reference herein: U.S. Nonprovisional patent application Ser. No. 17/463,161; filed Aug. 31, 2021; issued as U.S. U.S. Pat. No. 11,651,665 on May 16, 2023; and entitled DOORBELL COMMUNITIES. [0003] The entire contents of the following application are incorporated by reference herein: U.S. Nonprovisional patent application Ser. No. 16/863,893; filed Apr. 30, 2020; issued as U.S. U.S. Pat. No. 11,132,877 on Sep. 28, 2021; and entitled DOORBELL COMMUNITIES. [0004] The entire contents of the following application are incorporated by reference herein: U.S. Provisional Patent Application No. 62/143,032, filed Apr. 4, 2015, and entitled DOORBELL COMMUNITIES. [0005] The entire contents of the following applications are incorporated by reference herein: U.S. Nonprovisional patent application Ser. No. 14/612,376, filed Feb. 3, 2015, and entitled

DOORBELL COMMUNICATION SYSTEMS AND METHODS; U.S. Nonprovisional patent application Ser. No. 14/502,601, filed Sep. 30, 2014, and entitled DOORBELL COMMUNICATION SYSTEMS AND METHODS; U.S. Nonprovisional patent Application Ser. No. 14/492,809, filed Sep. 22, 2014, and entitled DOORBELL COMMUNICATION SYSTEMS AND METHODS; U.S. Nonprovisional patent application Ser. No. 14/275,811, filed May 12, 2014, and entitled DOORBELL COMMUNICATION SYSTEMS AND METHODS; U.S. Nonprovisional patent application Ser. No. 14/142,839, filed Dec. 28, 2013, and entitled DOORBELL COMMUNICATION SYSTEMS AND METHODS; U.S. Nonprovisional patent application Ser. No. 14/099,888, filed Dec. 6, 2013, and entitled DOORBELL COMMUNICATION SYSTEMS AND METHODS; and U.S. Nonprovisional patent application Ser. No. 14/098,772, filed Dec. 6, 2013, and entitled DOORBELL COMMUNICATION SYSTEMS AND METHODS. [0006] The entire contents of the following application are incorporated by reference herein: International Application No. PCT/US14/47622, filed Jul. 22, 2014 with the U.S. Patent and Trademark Office and entitled DOORBELL COMMUNICATION SYSTEMS AND METHODS. [0007] The entire contents of the following application are incorporated by reference herein: International Application No. PCT/US14/53506, filed Aug. 29, 2014 with the U.S. Patent and Trademark Office, and entitled DOORBELL COMMUNICATION SYSTEMS AND METHODS.

BACKGROUND

Field

[0008] Various embodiments disclosed herein relate to doorbells. Certain embodiments relate to doorbell communities.

Description of Related Art

[0009] Doorbells can enable a person located outside of an entry point, such as a door, to alert a person inside of an entry point that someone outside would like to talk to someone inside. Doorbells sometimes include a button located near a door, such as a front door, side door, or back door of a home, office, dwelling, warehouse, building, or structure. Doorbells are sometimes used near a gate or some other entrance to a partially enclosed area. Pushing the doorbell sometimes causes a chime or other alerting sound to be emitted.

SUMMARY

[0010] Data from multiple doorbells can be shared among a doorbell user group. Information regarding a visitor to a first building can be shared with other members of the doorbell user group. For example, a homeowner can share information from her doorbell with neighbors' remote computing devices via the doorbell user group even though the neighbors' remote computing devices are not configured to receive visitor alerts from the homeowner's doorbell each time the homeowner's doorbell detects a visitor (e.g., via motion detection or button press detection). Some embodiments include the homeowner (e.g., a user) choosing to send a first visit information (regarding a first visitor) to neighbor's remote computing devices and choosing not to send a second visit information (regarding a second visitor) to neighbor's remote computing devices. Thus, the system enables a user to selectively decide what visit information to share with members of the doorbell user group.

[0011] In some embodiments, the doorbell system can comprise a first remote computing device communicatively coupled to a first doorbell having a first camera. The method for using the doorbell system can comprise sending a first picture, taken by the first doorbell, of a first visitor to the first remote computing device. A doorbell user group can be created and can comprise a first user, of the first remote computing device, and a second user, of a second remote computing device. A second remote computing device can be communicatively coupled to a second doorbell, but may not necessarily be communicatively coupled to the first doorbell. The method for using the doorbell system can comprise sharing the first picture of the first visitor with the doorbell user

group, such that the second remote computing device can receive the first picture.

[0012] In several embodiments, the first remote computing device may not be communicatively coupled to the second doorbell. The method can comprise receiving, by the first remote computing device, a second picture taken by the second doorbell. The second picture can be taken in response to the second doorbell sending the second picture to a database. The database can be configured to provide visitor information to the doorbell user group.

[0013] In some embodiments, the method for using the doorbell system can comprise the indication of a first trait of the first visitor via the first remote computing device. The method can comprise sharing the first picture of the first visitor with the doorbell user group in response to the first trait of the first visitor.

[0014] In several embodiments, the method can comprise the indication of a first trait of the first visitor via the first remote computing device. The method can send the first trait and the first picture of the first visitor to a database. The database can be configured to provide information regarding the first visitor to the doorbell user group. The method can provide a time, at which the first visitor visited the first doorbell, to the database.

[0015] In some embodiments, the first remote computing device cannot be communicatively coupled to the second doorbell. The method for using the doorbell system can comprise receiving, by the first remote computing device, data from the doorbell user group regarding the second doorbell. The second doorbell can comprise a second camera. In some cases, a third doorbell can comprise a third camera. The second doorbell can be coupled to a second building, and the third doorbell can be coupled to a third building.

[0016] In several embodiments, the doorbell system can comprise a database configured to share visitor information with the doorbell user group. The method can comprise selecting, via the first remote computing device, to receive the visitor information from at least one of the second doorbell and the third doorbell.

[0017] In some embodiments, the first doorbell can be coupled to a first building. The first building, the second building, and the third building can be located apart from each other. Receiving the data can comprise receiving the visitor information from a different location than a first location of the first building.

[0018] In several embodiments, the first picture can comprise a first video. The first picture can also be a still image (although many users prefer pictures that are videos).

[0019] In some embodiments, creating the doorbell user group can comprise adding a first email address and a second email address to a database. The first email address can be associated with the first doorbell, and the second email address can be associated with the second doorbell.

[0020] In several embodiments, the method for using the doorbell system can comprise displaying a map on the first remote computing device. The map can display a second location of the second doorbell and a third location of a third doorbell. The method can comprise displaying information, regarding the second doorbell, in response to selecting a map icon, representing the second doorbell, while the map icon is displayed on the first remote computing device. Selecting the map icon can be used to add the second user of the second doorbell to the doorbell user group. Methods can include sending an invitation (to join the user group) to the second user in response to selecting the second doorbell on the map.

[0021] In some embodiments, the method can comprise selecting a first button on the first remote computing device to alert law enforcement and/or to alert the doorbell user group.

[0022] In several embodiments, the first doorbell can be coupled to a first building. The second doorbell can be coupled to a second building. A third doorbell can be coupled to a third building. The buildings can be located remotely relative to each other. The method can comprise analyzing information collected from the first, second, and third doorbells, and then sending an alert in response to analytics based on the information. Analytics can include visitor identity, visit frequency (e.g., during a timeframe), time of visits, and facial recognition of visitors.

[0023] In some embodiments, a method of using a doorbell system can comprise coupling a first doorbell having a first camera to a first building, configuring the first doorbell to wirelessly communicate with a first remote computing device, and joining, by the first remote computing device, a doorbell user group. The doorbell user group can comprise a second user and a third user. The second user can comprise a second doorbell having a second camera coupled to a second building. A third user can comprise a third doorbell having a third camera coupled to a third building. The second doorbell can be configured to wirelessly communicate with a second remote computing device.

[0024] In several embodiments, the method of using the doorbell system can comprise receiving, by the first remote computing device, a first visitor picture. The second doorbell or the third doorbell can take the first visitor picture. The method can comprise receiving, from the doorbell user group by the first remote computing device, a first visitor picture taken by the second doorbell. The first remote computing device may not be communicatively coupled to the second doorbell.

[0025] In some embodiments, the method can comprise receiving, by the first remote computing device, a first trait. The first trait can be regarding a first visitor in response to the second user inputting the first trait into the second remote computing device.

[0026] In several embodiments, the method of using the doorbell system can comprise taking a second visitor picture, by the first doorbell, and sending the second visitor picture to the doorbell user group. The second remote computing device can display the second visitor picture, even though the second remote computing device may not be communicatively coupled to the first doorbell.

[0027] In some embodiments, the doorbell user group can comprise a database having a first visitor picture taken by the second doorbell of the second building. The method can comprise taking, by the first doorbell of the first building, a second visitor picture. The method can comprise determining, by a portion of the doorbell system, that the first visitor picture and the second visitor picture show a visitor.

[0028] In several embodiments, the method can comprise labeling the visitor with solicitor information. The members of the doorbell user group can then see the first visitor picture associated with the solicitor information. The method can comprise alerting the members regarding at least a portion of the solicitor information.

[0029] In some embodiments, the method can comprise receiving a solicitor alert. The alert can be in response to the second doorbell, of the second building, and the third doorbell, of the third building, detecting a visitor.

[0030] In several embodiments, the method can comprise adding solicitation information regarding the visitor to the doorbell user group.

[0031] In some embodiments, the method can comprise sending an alert to members of the doorbell user group in response to detecting, by a remote burglar detection system, a burglary of the first building.

[0032] In several embodiments, a doorbell system can comprise a first doorbell that can have a first camera that can be coupled to a first building. The first doorbell can be communicatively coupled to a first remote computing device. A second doorbell can have a second camera that can be coupled to a second building. The second doorbell can be communicatively coupled to a second remote computing device. The doorbell system can have a doorbell user group that can comprise a database that can have images taken by the first camera and the second camera. The doorbell user group can be communicatively coupled to the first remote computing device and the second remote computing device.

[0033] In some embodiments of the doorbell system, the first doorbell may not be communicatively coupled to the second remote computing device. The second doorbell may not be communicatively coupled to the first remote computing device. The system can be configured such that the first remote computing device can receive a first visitor picture taken by the second

doorbell via the doorbell user group.

[0034] In several embodiments, the doorbell system can comprise a second visitor picture taken by the first doorbell. The picture can be displayed by the second remote computing device.

[0035] In some embodiments, the doorbell system can comprise a remote sensor. The remote sensor can be configured to monitor the first building and can detect an unauthorized building intrusion. An alert can be sent to the doorbell user group in response to the remote sensor detecting the unauthorized building intrusion.

[0036] In several embodiments, the system can comprise a solicitor alert. A solicitor alert can be sent to the doorbell user group in response to a visitor being detected by the first doorbell, at the first building, and by the second doorbell, at the second building.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0037] These and other features, aspects, and advantages are described below with reference to the drawings, which are intended to illustrate, but not to limit, the invention. In the drawings, like reference characters denote corresponding features consistently throughout similar embodiments.

[0038] FIG. 1 illustrates a front view of a doorbell system, according to some embodiments. FIG. 2 illustrates a front view of a computing device running software to communicate with the doorbell from FIG. 1, according to some embodiments.

[0039] FIG. 3 illustrates a diagrammatic view of an embodiment in which the doorbell from FIG. 1 is connected to a building, according to some embodiments.

[0040] FIG. 4 illustrates a back view of the doorbell from FIG. 1 without a mounting bracket, according to some embodiments.

[0041] FIG. 5 illustrates a diagrammatic view of a group of doorbells, according to some embodiments.

[0042] FIG. 6 illustrates a front view of a remote computing device displaying a map, according to some embodiments.

[0043] FIG. 7 illustrates a diagrammatic view of picture being communicated, according to some embodiments.

[0044] FIG. 8 illustrates a diagrammatic view of a doorbell that is communicatively coupled to multiple remote computing devices, according to some embodiments.

[0045] FIG. 9 illustrates a diagrammatic view of a remote computing device that is communicatively coupled to multiple doorbells, according to some embodiments.

[0046] FIGS. 10 and 11 illustrate diagrammatic views of doorbell user groups, according to some embodiments.

DETAILED DESCRIPTION

[0047] Although certain embodiments and examples are disclosed below, inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses, and to modifications and equivalents thereof. Thus, the scope of the claims appended hereto is not limited by any of the particular embodiments described below. For example, in any method or process disclosed herein, the acts or operations of the method or process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various operations may be described as multiple discrete operations in turn, in a manner that may be helpful in understanding certain embodiments; however, the order of description should not be construed to imply that these operations are order dependent. Additionally, the structures, systems, and/or devices described herein may be embodied as integrated components or as separate components.

[0048] For purposes of comparing various embodiments, certain aspects and advantages of these

embodiments are described. Not necessarily all such aspects or advantages are achieved by any particular embodiment. Thus, for example, various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may also be taught or suggested herein.

System Embodiments

[0049] Communication systems can provide a secure and convenient way for a remotely located individual to communicate with a person who is approaching a sensor, such as a proximity sensor or motion sensor, or with a person who rings a doorbell, which can be located in a doorway, near an entrance, or within 15 feet of a door. Some communication systems allow an individual to hear, see, and talk with visitors who approach at least a portion of the communication system and/or press a button, such as a doorbell's button. For example, communication systems can use a computing device to enable a remotely located person to see, hear, and/or talk with visitors.

Computing devices can include computers, laptops, tablets, mobile devices, smartphones, cellular phones, and wireless devices (e.g., cars with wireless communication). Example computing devices include the iPhone, iPad, iMac, MacBook Air, and MacBook Pro made by Apple Inc.

Communication between a remotely located person and a visitor can occur via the Internet, cellular networks, telecommunication networks, and wireless networks.

[0050] Referring now to FIG. 1, doorbell systems can be a portion of a smart home hub. In some embodiments, the doorbell system **200** forms the core of the smart home hub. For example, the various systems described herein enable complete home automation. In some embodiments, the doorbell **202** controls various electrical items in a home (e.g., lights, air conditioners, heaters, motion sensors, garage door openers, locks, televisions, computers, entertainment systems, pool monitors, elderly monitors). In some embodiments, the computing device **204** controls the doorbell **202** and other electrical items in a home (e.g., lights, air conditioners, heaters, motion sensors, garage door openers, locks, televisions, computers, entertainment systems, pool monitors, elderly monitors).

[0051] FIG. 1 illustrates a front view of a communication system embodiment. The doorbell system **200** can include a doorbell **202** (e.g., a security system) and a computing device **204**. Although the illustrated doorbell **202** includes many components in one housing, several doorbell embodiments include components in separate housings. The doorbell **202** can include a camera assembly **208** and a doorbell button **212**. The camera assembly **208** can include a video camera, which in some embodiments is a webcam. The doorbell **202** can include a diagnostic light **216** and a power indicator light **220**. In some embodiments, the diagnostic light **216** is a first color (e.g., blue) if the doorbell **202** and/or the doorbell system **200** is connected to a wireless Internet network and is a second color (e.g., red) if the doorbell **202** and/or the doorbell system **200** is not connected to a wireless Internet network. In some embodiments, the power indicator **220** is a first color if the doorbell **202** is connected to a power source. The power source can be supplied by the building to which the doorbell **202** is attached. In some embodiments, the power indicator **220** is a second color or does not emit light if the doorbell **202** is not connected to the power source.

[0052] The doorbell **202** can include an outer housing **224**, which can be water resistant and/or waterproof. The outer housing can be made from metal or plastic, such as molded plastic with a hardness of 60 Shore D. In some embodiments, the outer housing **224** is made from brushed nickel or aluminum.

[0053] Rubber seals can be used to make the outer housing **224** water resistant or waterproof. The doorbell **202** can be electrically coupled to a power source, such as wires electrically connected to a building's electrical power system. In some embodiments, the doorbell **202** includes a battery for backup and/or primary power.

[0054] Wireless communication **230** can enable the doorbell **202** to communicate with the computing device **204**. Some embodiments enable communication via cellular and/or WiFi networks. Some embodiments enable communication via the Internet. Several embodiments enable

wired communication between the doorbell **202** and the computing device **204**. The wireless communication **230** can include the following communication means: radio, WiFi (e.g., wireless local area network), cellular, Internet, Bluetooth, telecommunication, electromagnetic, infrared, light, sonic, and microwave. Other communication means are used by some embodiments. In some embodiments, such as embodiments that include telecommunication or cellular communication means, the doorbell **202** can initiate voice calls or send text messages to a computing device **204** (e.g., a smartphone, a desktop computer, a tablet computer, a laptop computer).

[0055] Several embodiments use near field communication (NFC) to communicate between the computing device **204** and the doorbell **202**. The doorbell **202** and/or the computing device **204** can include a NFC tag. Some NFC technologies include Bluetooth, radio-frequency identification, and QR codes.

[0056] Some embodiments include computer software (e.g., application software), which can be a mobile application designed to run on smartphones, tablet computers, and other mobile devices. Software of this nature is sometimes referred to as “app” software. Some embodiments include software designed to run on desktop computers and laptop computers.

[0057] The computing device **204** can run software with a graphical user interface. The user interface can include icons or buttons. In some embodiments, the software is configured for use with a touch-screen computing device such as a smartphone or tablet.

[0058] FIG. **2** illustrates a computing device **204** running software. The software includes a user interface **240** displayed on a display screen **242**. The user interface **240** can include a doorbell indicator **244**, which can indicate the location of the doorbell that the user interface is displaying. For example, a person can use one computing device **204** to control and/or interact with multiple doorbells, such as one doorbell located at a front door and another doorbell located at a back door. Selecting the doorbell indicator **244** can allow the user to choose another doorbell (e.g., the back door's doorbell rather than the front door's doorbell).

[0059] The user interface **240** can include a connectivity indicator **248**. In some embodiments, the connectivity indicator can indicate whether the computing device is in communication with a doorbell, the Internet, and/or a cellular network. The connectivity indicator **248** can alert the user if the computing device **204** has lost its connection with the doorbell **202**; the doorbell **202** has been damaged; the doorbell **202** has been stolen; the doorbell **202** has been removed from its mounting location; the doorbell **202** has lost electrical power; and/or if the computing device **204** cannot communicate with the doorbell **202**. In some embodiments, the connectivity indicator **248** alerts the user of the computing device **204** by flashing, emitting a sound, displaying a message, and/or displaying a symbol.

[0060] In some embodiments, if the doorbell **202** loses power, loses connectivity to the computing device **204**, loses connectivity to the Internet, and/or loses connectivity to a remote server, a remote server **206** sends an alert (e.g., phone call, text message, image on the user interface **240**) regarding the power and/or connectivity issue. In several embodiments, the remote server **206** can manage communication between the doorbell **202** and the computing device. In some embodiments, information from the doorbell **202** is stored by the remote server **206**. In several embodiments, information from the doorbell **202** is stored by the remote server **206** until the information can be sent to the computing device **204**, uploaded to the computing device **204**, and/or displayed to the remotely located person via the computing device **204**. The remote server **206** can be a computing device that stores information from the doorbell **202** and/or from the computing device **204**. In some embodiments, the remote server **206** is located in a data center.

[0061] In some embodiments, the computing device **204** and/or the remote server **206** attempts to communicate with the doorbell **202**. If the computing device **204** and/or the remote server **206** is unable to communicate with the doorbell **202**, the computing device **204** and/or the remote server **206** alerts the remotely located person via the software, phone, text, a displayed message, and/or a website. In some embodiments, the computing device **204** and/or the remote server **206** attempts to

communicate with the doorbell **202** periodically; at least every five hours and/or less than every 10 minutes; at least every 24 hours and/or less than every 60 minutes; or at least every hour and/or less than every second.

[0062] In some embodiments, the server **206** can initiate communication with the computing device **204** and/or with the doorbell **202**. In several embodiments, the server **206** can initiate, control, and/or block communication between the computing device **204** and the doorbell **202**.

[0063] In several embodiments, a user can log in to an “app,” website, and/or software on a computing device (e.g., mobile computing device, smartphone, tablet, desktop computer) to adjust the doorbell settings discussed herein.

[0064] In some embodiments, a computing device can enable a user to watch live video and/or hear live audio from a doorbell due to the user's request rather than due to actions of a visitor. Some embodiments include a computing device initiating a live video feed (or a video feed that is less than five minutes old).

[0065] In some embodiments, the user interface **240** displays an image **252** such as a still image or a video of an area near and/or in front of the doorbell **202**. The image **252** can be taken by the camera assembly **208** and stored by the doorbell **202**, server **206**, and/or computing device **204**. The user interface **240** can include a recording button **256** to enable a user to record images, videos, and/or sound from the camera assembly **208**, microphone of the doorbell **202**, and/or microphone of the computing device **204**.

[0066] In several embodiments, the user interface **240** includes a picture button **260** to allow the user to take still pictures and/or videos of the area near and/or in front of the doorbell **202**. The user interface **240** can also include a sound adjustment button **264** and a mute button **268**. The user interface **240** can include camera manipulation buttons such as zoom, pan, and light adjustment buttons. In some embodiments, the camera assembly **208** automatically adjusts between Day Mode and Night Mode. Some embodiments include an infrared camera and/or infrared lights to illuminate an area near the doorbell **202** to enable the camera assembly **208** to provide sufficient visibility (even at night).

[0067] In some embodiments, buttons include diverse means of selecting various options, features, and functions. Buttons can be selected by mouse clicks, keyboard commands, or touching a touch screen. Many embodiments include buttons that can be selected without touch screens.

[0068] In some embodiments, the user interface **240** includes a quality selection button, which can allow a user to select the quality and/or amount of the data transmitted from the doorbell **202** to the computing device **204** and/or from the computing device **204** to the doorbell **202**.

[0069] In some embodiments, video can be sent to and/or received from the computing device **204** using video chat protocols such as FaceTime (by Apple Inc.) or Skype (by Microsoft Corporation). In some embodiments, these videos are played by videoconferencing apps on the computing device **204** instead of being played by the user interface **240**.

[0070] The user interface **240** can include a termination button **276** to end communication between the doorbell **202** and the computing device **204**. In some embodiments, the termination button **276** ends the ability of the person located near the doorbell **202** (i.e., the visitor) to hear and/or see the user of the computing device **204**, but does not end the ability of the user of the computing device **204** to hear and/or see the person located near the doorbell **202**.

[0071] In some embodiments, a button **276** is both an answer button (to accept a communication request from a visitor) and a termination button (to end communication between the doorbell **202** and the computing device **204**). The button **276** can include the word “Answer” when the system is attempting to establish two-way communication between the visitor and the user. Selecting the button **276** when the system is attempting to establish two-way communication between the visitor and the user can start two-way communication. The button **276** can include the words “End Call” during two-way communication between the visitor and the user. Selecting the button **276** during two-way communication between the visitor and the user can terminate two-way communication.

In some embodiments, terminating two-way communication still enables the user to see and hear the visitor. In some embodiments, terminating two-way communication causes the computing device **204** to stop showing video from the doorbell and to stop emitting sounds recorded by the doorbell.

[0072] In some embodiments, the user interface **240** opens as soon as the doorbell detects a visitor (e.g., senses indications of a visitor). Once the user interface **240** opens, the user can see and/or hear the visitor even before “answering” or otherwise accepting two-way communication, in several embodiments.

[0073] Some method embodiments include detecting a visitor with a doorbell. The methods can include causing the user interface to display on a remote computing device **204** due to the detection of the visitor (e.g., with or without user interaction). The methods can include displaying video from the doorbell and/or audio from the doorbell before the user accepts two-way communication with the visitor. The methods can include displaying video from the doorbell and/or audio from the doorbell before the user accepts the visitor's communication request. The methods can include the computing device simultaneously asking the user if the user wants to accept (e.g., answer) the communication request and displaying audio and/or video of the visitor. For example, in some embodiments, the user can see and hear the visitor via the doorbell before opening a means of two-way communication with the visitor.

[0074] In some embodiments, the software includes means to start the video feed on demand. For example, a user of the computing device might wonder what is happening near the doorbell **202**. The user can open the software application on the computing device **204** and instruct the application to show live video and/or audio from the security device **202** even if no event near the doorbell **202** has triggered the communication.

[0075] In several embodiments, the security device **202** can be configured to record when the security device **202** detects movement and/or the presence of a person. The user of the computing device **204** can later review all video and/or audio records from when the security device **202** detected movement and/or the presence of a person.

[0076] Referring now to FIG. **1**, in some embodiments, the server **206** controls communication between the computing device **204** and the doorbell **202**, which can be a doorbell with a camera, a microphone, and a speaker. In several embodiments, the server **206** does not control communication between the computing device **204** and the doorbell **202**.

[0077] In some embodiments, data captured by the doorbell and/or the computing device **204** (such as videos, pictures, and audio) is stored by another remote device such as the server **206**. Cloud storage, enterprise storage, and/or networked enterprise storage can be used to store video, pictures, and/or audio from the doorbell system **200** or from any part of the doorbell system **200**. The user can download and/or stream stored data and/or storage video, pictures, and/or audio. For example, a user can record visitors for a year and then later can review conversations with visitors from the last year. In some embodiments, remote storage, the server **206**, the computing device **204**, and/or the doorbell **202** can store information and statistics regarding visitors and usage.

[0078] FIG. **3** illustrates an embodiment in which a doorbell **202** is connected to a building **300**, which can include an entryway **310** that has a door **254**. Electrical wires **304** can electrically couple the doorbell **202** to the electrical system of the building **300** such that the doorbell **202** can receive electrical power from the building **300**. The building can include a door lock **250** to lock the door **254**.

[0079] A wireless network **308** can allow devices to wirelessly access the Internet. The doorbell **202** can access the Internet via the wireless network **308**. The wireless network **308** can transmit data from the doorbell **202** to the Internet, which can transmit the data to remotely located computing devices **204**. The Internet and wireless networks can transmit data from remotely located computing devices **204** to the doorbell **202**. In some embodiments, a doorbell **202** connects to a home's WiFi.

[0080] As illustrated in FIG. 3, one computing device **204** (e.g., a laptop, a smartphone, a mobile computing device, a television) can communicate with multiple doorbells **202**. In some embodiments, multiple computing devices **204** can communicate with one doorbell **202**.

[0081] In some embodiments, the doorbell **202** can communicate (e.g., wirelessly **230**) with a television **306**, which can be a smart television. Users can view the television **306** to see a visitor and/or talk with the visitor.

[0082] FIG. 4 illustrates an internal view of the doorbell **202**. Doorbells **202** can include a chip **480** (e.g., integrated circuits, microprocessor, computer) and a memory **492**. Doorbells **202** can also include a microphone **484** and a speaker **488**. The speaker **488** can comprise a flat speaker and a sound chamber **460** configured to amplify an emitted sound. The flat speaker can be located in the sound chamber. Some doorbell embodiments include a proximity sensor **500**. In several embodiments, doorbells **202** include a wireless communication module **504**, such as a WiFi module. The communication module **504** can have an integrated antenna. In some embodiments, an antenna is contained within the outer housing **224**.

[0083] The doorbell **202** can include one or more heating elements **508** configured to regulate the temperature of the doorbell **202**. For example, doorbells **202** can be used in very cold environments, such as in Alaska. The heating element **508** can be used in various methods to protect temperature sensitive portions of the doorbell **202** from cold weather.

[0084] While protecting the doorbell **202** from cold weather can be important in some embodiments, protecting visitors from excessive heat can also be important in some embodiments. Excessive heat could burn visitors as they “ring” the doorbell (e.g., press the doorbell button **212** shown in FIG. 10). The doorbell **202** can include a thermometer **512** to enable the system to determine the temperature inside a portion of the doorbell **202** and/or outside the doorbell **202**.

[0085] Several embodiments can be configured for 9 to 40 volts alternating current (“VAC”) and/or 9 to 40 volts direct current (“VDC”). Some embodiments convert input electricity into direct current (DC), such as 12 VDC. Several embodiments include a converter **494** for power conversion (e.g., converting electrical energy from one form to another). The converter **494** can convert input power (e.g., from wiring in a building) to a suitable power form for the doorbell **202**. The power conversion can convert between AC and DC, change the voltage, and/or change the frequency. The converter **494** can include a transformer and/or a voltage regulator. In several embodiments, the converter **494** can include an AC to DC converter, a DC to DC converter, a voltage stabilizer, a linear regulator, a surge protector, a rectifier, a power supply unit, a switch, an inverter, and/or a voltage converter. In some embodiments, the converter **494** converts 50 Hertz (“Hz”) power into 60 Hz power.

[0086] The electrical components of the doorbell **202** (e.g., the camera assembly **208**, the memory **492**, the chip **480**, the speaker **488**, the converter **494**, the microphone **484**, the lights **458**, the rectifier **524**, the proximity sensor **500**, the communication module **504**, the heating element **508**, the electrical connectors **510**, the thermometer **512**, the image analysis system **520**, and the battery **642**) can be electrically coupled to a printed circuit board (“PCB”) **516** and can receive electrical power from the PCB **516**.

[0087] The PCB **516** and the electrical components of the doorbell **202** can be the electrical system **456** of the doorbell **202**. Additional details regarding the PCB **516** and the electrical components of the doorbell **202** are described in U.S. Nonprovisional patent Application Ser. No. 14/612,376; filed Feb. 3, 2015; and entitled DOORBELL COMMUNICATION SYSTEMS AND METHODS. The entire contents of patent application Ser. No. 14/612,376 are incorporated by reference herein.

[0088] Although some embodiments are described in the context of methods, the method embodiments can also be formulated as devices and systems. Methods described herein can be applied to the devices and systems incorporated by references herein.

Systems with Multiple Doorbells

[0089] Multiple doorbells can share data to “watch” over a neighborhood by letting people connect

with neighbors and share information regarding suspicious visitors. The shared information can include pictures of visitors and the time of suspicious activity. As a result, sharing data between multiple doorbell users can help reduce crime and theft in neighborhoods around the world.

[0090] Suspicious activity can be “tagged” and shared with members of a user group (e.g., a group of neighbors). The shared data can include photos and information regarding suspicious visitors. The information can include written descriptions, recorded descriptions, and videos. For example, a first doorbell can take a picture of a visitor. The doorbell can send the picture to a remote computing device. A user of the remote computing device can see the visitor. Then, the user can enter information regarding the suspicious visitor into the remote computing device. The doorbell system can then share the information and the picture of the visitor with a user group, which can include other doorbell users.

[0091] An administrator of the user group can be a “master user” who has the ability to add other users to the user group. One way of adding users to a user group is to enter email addresses of prospective users into a remote computing device (e.g., into an “app” running on the remote computing device). This app can be the same app used by the administrator to configure her doorbell.

[0092] In a user group section (e.g., a “neighborhood section”) of the app, a user can see who else is in her neighborhood. In some embodiments, the home locations of prospective or current users are displayed on a map. Selecting prospective users can enable the administrator to add new users to the user group.

[0093] When a user thinks a visitor is suspicious, the user can report the visitor and the nature of the suspicious activity to the user group. Thus, the user can enter the visitor into an activity log such that the event has a “suspicious” tag. The event can then be shared with the other users via email, text message, a push notifications, or any suitable means. The event can be recorded in a user group activity list, which in some embodiments, is only visible to members of the user group (or at least is not visible to the general public). The event can also be listed in a history of events reported by a specific individual. Thus, users can see events they reported, and users can see events reported by all members of a user group.

[0094] Each event can include an icon. Selecting the icon can cause the event to be shared with the user group. In some embodiments, icons are shown on a map that is displayed on a remote computing device. In this manner, users can see where suspicious visitors have been reported by members of the user group, and in some embodiments, by members of other user groups.

[0095] In some embodiments, a user group is limited to a particular neighborhood. In other embodiments, user groups are not limited to a particular neighborhood.

[0096] Visitor data from a user (or from a user group) can be shared with law enforcement (e.g., police officers) or other safety personnel to help keep neighborhoods safe. A suspicious event reported by a first user can be shared with a second user. The second user can then share the event with law enforcement. In some embodiments, the user who reports the event to the user group also reports the event to law enforcement (e.g., by selecting a button in the app).

[0097] The app can include a “distress button” configured to enable a user to alert law enforcement and user group members that something is wrong. Pressing the button on a remote computing device can cause a doorbell system to send a push notification to other users to notify them that help is needed. The push notification can also communicate the nature of the event and can enable users to see a picture of the visitor causing the trouble.

[0098] FIG. 5 illustrates a neighborhood, which can include many buildings **300a**, **300b**, **300c**. The buildings **300a**, **300b**, **300c** can be individual homes or individual buildings (e.g., multiple single-family homes, multiple apartments coupled together, offices). The buildings **300a**, **300b**, **300c** can be located next to each other or they can be separated by other homes or buildings.

[0099] Embodiments can include diverse ways of using a doorbell system **386**, which can include multiple doorbells **202a**, **202b**, **202c** and multiple remote computing devices **204a**, **204b**, **204c**.

Each doorbell **202a**, **202b**, **202c** can be coupled to an exterior wall of a different building **300a**, **300b**, **300c**. A different remote computing device **204a**, **204b**, **204c** can configure each doorbell **202a**, **202b**, **202c**. Each building **300a**, **300b**, **300c** can have its own wireless network **308a**, **308b**, **308c**. Another network **308d** (e.g., a cellular network, the Internet) can enable a first doorbell **202a** to send a picture of a visitor **388a** to a second doorbell **202b** and/or to a second remote computing device **204b**.

[0100] A first doorbell **202a** can send a visitor notification (e.g., an alert as shown in FIG. 1) to a first computing device **204a** via a first wireless network **308a**. In some embodiments, however, it is advantageous for a doorbell **202a** or a remote computing device **204a** associated with a first building **300a** to send information regarding a visitor **388a** to a doorbell **202b** or a remote computing device **204b** associated with a second building **300b**. In several embodiments, it is advantageous for a doorbell **202a** or a remote computing device **204a** associated with a first building **300a** to receive information regarding a visitor **388b** from a doorbell **202b** or a remote computing device **204b** associated with a second building **300b**. An external network **308d** can facilitate this sending and receiving of information regarding visitors **388a**, **388b**. This way, the doorbell system **386** can enable notifying a second remote computing device **204b** regarding a visitor detected by a first doorbell **202a** even though the second remote computing device **204b** is not configured to control the first doorbell **202a**.

[0101] Each of the doorbells **202a**, **202b**, **202c** illustrated in FIG. 5 can include a camera assembly **208** (as labeled in FIG. 1). Some embodiments of using the doorbell system **386** include taking a first picture of a first visitor **388a** with the first doorbell **202a**; sending the first picture to the first remote computing device **204a**; and indicating a first trait (e.g., identity, suspicious behavior, bodily characteristics) of the first visitor **388a** via the first remote computing device **204a**.

[0102] The first remote computing device **204a** can be used to categorize the visitor. For example, the user of the first remote computing device can indicate whether the first visitor **388a** is a salesperson, a fundraiser, or a potential criminal. The first remote computing device **204a** can share this categorization with other members of the user group, which can include users of the remote computing devices **204a**, **204b**, **204c**.

[0103] A first user of the first remote computing device **204a** (e.g., the owner of the first doorbell **202a**) can create a user group. The user group can include the first user of the first remote computing device **204a** and a second user of a second remote computing device **204b**. Note that in FIG. 5, the first remote computing device **204a** is located remotely relative to the first doorbell **202a** because the first remote computing device is not electrically or mechanically coupled to the first doorbell **202a** (even though the first remote computing device **204a** and the first doorbell **202a** are located in the same building **300a**).

[0104] Embodiments can also include sharing the first picture of the first visitor **388a** with the user group. The sharing of the first picture of the first visitor **388a** with the user group can be in response to the first trait of the first visitor **388a**. For example, the first trait can be that the first visitor **388a** is acting suspiciously by peaking over fences or looking through windows.

[0105] Several embodiments include sending the first trait and the first picture of the first visitor **388a** to a database **390** configured to provide information regarding the first visitor **388a** to the user group. The first doorbell **202a** can also determine a time at which the first visitor **388a** visited the first doorbell **202a**. A portion of the doorbell system **386** can provide this time to the database **390**.

[0106] Some embodiments include receiving data with the first remote computing device **204a** regarding a second doorbell **202b** that comprises a second camera and regarding a third doorbell **202c** that comprises a third camera. As illustrated in FIG. 5, the first doorbell **202a** is coupled to a first building **300a**. The second doorbell **202b** is coupled to a second building **300b**. The third doorbell **202c** is coupled to a third building **300c**. Doorbells can be mounted near a door in an entryway. The buildings **300a**, **300b**, and **300c** can be located apart from each other such that each

building is a separate residential dwelling or a separate office space. The database **390** can be configured to share visitor information with the user group. Embodiments can include selecting (via the first remote computing device **204a**) to receive the visitor information from at least one of the second doorbell **202b** and the third doorbell **202c**. Receiving the data can comprise receiving the visitor information from a different location than a first location of the first building **300a**. [0107] Each doorbell **202a**, **202b**, and **202c** can include a camera assembly **208** (labeled in FIG. 1). In several embodiments, a camera assembly **208** can take videos or stationary images. Thus, a “picture” taken by a camera of a doorbell can actually be a video. In some embodiments, the “picture” is a stationary image (rather than a video).

[0108] There are many ways to create user groups. In some embodiments, creating a user group comprises adding a first email address and a second email address to a database. The first email address can be associated with the first doorbell **202a**. The second email address can be associated with a second doorbell **202b**. In some embodiments, a first user of a first doorbell **202a** sends an invitation to join a doorbell user group to a second user of a second doorbell **202b**. The first user can send the invitation via a first remote computing device **204a** (that is authorized to configure the first doorbell **202a**) to a second remote computing device **204b** (that is authorized to configure the second doorbell **202b**). The second user can then accept or decline the invitation via the second remote computing device **204b**. In some embodiments, the second user accepts or declines the invitation to join the doorbell user group at least partially in response to seeing the location of the first doorbell **202a** on the display of the second computing device **204b**. For example, if the first and second users are neighbors, then the second user could be more likely to accept the invitation than if the first and second users live much farther away from each other.

[0109] FIG. 6 illustrates a map **392** that is displayed by the first remote computing device **204a**. The map **392** displays a first location of the first doorbell **202a**, a second location of the second doorbell **202b**, and a third location of the third doorbell **202c**. The map **392** can also include other items such as street names. Several embodiments include displaying information regarding the second doorbell **202b** in response to a person using the first computing device **204a** to select a second icon representing the second doorbell **202b** while the second icon is displayed on the first remote computing device **204a**. (In FIG. 6, doorbells **202a**, **202b**, **202c** are represented by icons that look like doorbells.) This information can include an address of the second doorbell **202b**, contact information (e.g., a name, a phone number, an email) of a user of the second doorbell **202b**, and/or a picture of the user of the second doorbell **202b**. Selecting the second icon can add a second user of the second doorbell **202b** to the doorbell user group.

[0110] Not all members of a doorbell user group necessarily have a doorbell. Members of the user group who have not yet purchased a doorbell with a camera can still receive visitor information, including visitor pictures, on their remote computing devices. For example, a picture taken by the first doorbell **202a** can be sent to the remote computing devices of people who have not purchased a doorbell with a camera.

[0111] Several embodiments include selecting a first button **394** on the display of the first remote computing device **204a** to alert law enforcement and/or to alert the user group. The button **394** can be a portion of the graphical user interface of an app.

[0112] A first user can also join a user group that was previously created by another user. This first user can couple a first doorbell having a first camera to a first building; configure the first doorbell to wirelessly communicate with a first remote computing device; and/or join a user group via the first remote computing device. This user group can comprise a second user of a second doorbell having a second camera and a third user of a third doorbell having a third camera. The second doorbell can be configured to wirelessly communicate with a second remote computing device. This first user can receive a first picture and a first trait of a first visitor with the first remote computing device. This first picture could have been taken previously by the second doorbell or the third doorbell. This first user can take a second picture of a second visitor with the first doorbell;

and/or send the second picture of the second visitor to the user group such that the second remote computing device can display the second picture. This first user can detect a second visitor with the first doorbell. This first user can determine that the second visitor is the first visitor by a portion of the doorbell system analyzing the first picture from the second doorbell or the third doorbell. This analysis can include the doorbell system comparing the first picture to the second picture. Embodiments can also include alerting a first user of the first doorbell that the second visitor is the first visitor (i.e., the first visitor is the same person as the second visitor).

[0113] FIG. 7 illustrates that a first doorbell **202a** can take a picture **398**. Then, the first doorbell **202a** can send the picture **398** to a first remote computing device **204a**. (The doorbell **202a** can send the picture **398** directly to the first remote computing device **204a** or indirectly to the first remote computing device **204a** via an intermediary device such as a router, server, network, and/or the Internet.) Then, the first remote computing device **204a** can send the picture **398** to a second remote computing device **204b**. This approach enables the second remote computing device **204b** to receive the picture **398** even though the second remote computing device is not communicatively coupled to the doorbell **202a**.

[0114] In contrast, FIG. 8 illustrates an embodiment in which multiple remote computing devices **204a**, **204b**, **204c** are communicatively coupled with one doorbell **202a**. Thus, the doorbell **202a** can send a picture that it takes to all of the remote computing devices **204a**, **204b**, **204c**. In some cases, a first user might not want her doorbell to be communicatively coupled with a neighbor's remote computing device (e.g., because communicatively coupling her doorbell with her neighbor's remote computing device could enable the neighbor to see visitors and/or talk with visitors to the home of the first user without the first user's permission). As a result, the first user might want to be able to share a visitor picture and/or visitor information with a neighbor without the neighbor's remote computing device being communicatively coupled with the first user's doorbell. The first user's remote computing device, a user group, a server and/or a database can act as a gatekeeper that enables the first user to control what visitor information she shares with other members of a user group (e.g., with neighbors or members of other households).

[0115] In some embodiments, a homeowner can share information from her doorbell with neighbors' remote computing devices via the doorbell user group **400** (shown in FIG. 10) even though the neighbors' remote computing devices are not configured to receive visitor alerts from the homeowner's doorbell each time the homeowner's doorbell detects a visitor (e.g., via motion detection or button press detection). Some embodiments include the homeowner (e.g., a user) choosing to send a first visit information (regarding a first visitor) to neighbor's remote computing devices and choosing not to send a second visit information (regarding a second visitor) to neighbor's remote computing devices. This choosing step can be after and in response to seeing (e.g., displaying) the visitors on a remote computing device. Thus, the system **408** (shown in FIG. 10) enables a user to selectively decide what visit information to share with members of the doorbell user group.

[0116] In contrast, the configuration illustrated in FIG. 8 does not enable selectively deciding what visit information to share. Instead, all the remote computing devices **204a**, **204b**, **204c** receive the visit information. While the configuration illustrated in FIG. 8 may work well for one household, it typically does not work well with multiple households (e.g., an entire neighborhood) due to privacy concerns and due to the fact that most neighbors would not want to receive a notification every time a person approaches any home in the neighborhood. The number of unhelpful notifications would be very bothersome. Imagine how many uneventful notifications a user would receive each day after school as dozens of children come home. Thus, the system **408** illustrated in FIG. 10 works much better than the system illustrated in FIG. 8 when multiple households, homes, and/or buildings are involved in the visitor information sharing.

[0117] For example, as shown in FIG. 11, the remote computing device **204b** and/or the user group **400** can act as a gatekeeper system to enable a user to select which visitor information is shared

other remote computing devices **204a**, **204c** after the user's remote computing device **204b** receives the visitor information. This approach allows a user to share (e.g., by selecting a button on the device **204b**) information regarding some visitors (e.g., suspicious visitors) with other remote computing devices **204a**, **204c** after seeing (or displaying) the visitors on the computing device **204b**. This approach also allows the user to not share information regarding other visitors (e.g., trusted friends) with other remote computing devices **204a**, **204c** after seeing (or displaying) the other visitors on the computing device **204b**. Thus, the “share decision” can be in response to the user seeing (e.g., analyzing) the visitors on her remote computing device **204b**.

[0118] In contrast, the system shown in FIG. **8** does not include a visitor by visitor “share decision.” Instead, a user simply adds several computing devices **204a**, **204b**, **204c** to a system such that the computing devices **204a**, **204b**, **204c** receive a visitor notification (e.g., an alert) each time the doorbell **202a** detects a visitor in response to the computing devices **204a**, **204b**, **204c** being communicatively coupled with the doorbell **202a**.

[0119] FIG. **9** illustrates an embodiment in which a remote computing device **204a** is communicatively coupled (e.g., wirelessly) with multiple doorbells **202a**, **202b**, **202c**. For example, a first user could be able to see and/or talk with visitors detected by her own doorbell **202a** or detected by neighbors' doorbells **202b**, **202c**. This approach raises privacy concerns.

[0120] FIG. **10** illustrates a diagrammatic view of an embodiment that includes a user group **400**. The user group **400** can include a database **390**, a server **404**, and user information **406** such as email addresses, user profiles, and user doorbell locations.

[0121] Thus, many embodiments enable members of a user group to share visitor information with each other without granting doorbell control to members of the user group. For example, a first user can alter at least one setting (e.g., a chime tone, a greeting, a silent mode, on-demand video modes) of her doorbell but typically cannot alter settings of neighbors' doorbells (e.g., of other doorbells in the user group). These embodiments enable a first user to determine what visitor pictures and information her doorbell shares with a user group.

[0122] The doorbell system **408** can include a first remote computing device **204a** communicatively coupled to a first doorbell **202a** that has a first camera **208** (labeled in FIG. **1**). Some embodiments include sending a first picture **410**, taken by the first doorbell **202a**, of a first visitor **402a** to the first remote computing device **204a**. An arrow indicates the first doorbell **202a** is communicatively coupled with the first remote computing device **204a**. This communicatively coupling can be wireless and can include two-way communication to enable the first remote computing device **204a** to send data and doorbell setting parameters to the first doorbell **202a**.

[0123] The first doorbell **202a** is not communicatively coupled to the second remote computing device **204b** even though the second remote computing device **204b** can receive a picture taken by the first doorbell **202a** from at least a portion of the user group **400**. This way, the second remote computing device **204b** can receive information regarding a visitor detected by the first doorbell **202a**, but the second remote computing device **204b** cannot alter settings of the first doorbell **202a**, initiate on-demand videos from the first doorbell **202a**, or talk with the first visitor **402a** via the first doorbell **202a**.

[0124] Embodiments can include creating a doorbell user group **400** comprising a first user of the first remote computing device **204a** and a second user of the second remote computing device **204b**, which is communicatively coupled to a second doorbell **202b** (as shown by the arrow) but is not communicatively coupled to the first doorbell **202a**. The remote computing devices **204a**, **204b**, **204c** can be communicatively coupled with the user group **400** to enable sharing visitor information (e.g., pictures, audio files, visitor descriptions) detected by one of the doorbells **202a**, **202b**, **202c** (and/or recorded by one of the remote computing devices **204a**, **204b**, **204c**) with all of the remote computing devices **204a**, **204b**, **204c** communicatively coupled with the user group **400**. This way, visitor information can be shared with members of the user group **400** without users being concerned that other members of the user group **400** might hear private visitor conversations

and/or see confidential visitors. Embodiments can enable a user to control what visitor information members of the user group receive from the user's doorbell.

[0125] Embodiments can include sharing the first picture **410** of the first visitor **402a** with the doorbell user group **400** such that the second remote computing device **204b** receives the first picture **410** taken by the first doorbell **202a**. Sharing the first picture **410** with the doorbell user group **400** can require the user of the first remote computing device **204a** to authorize sharing the first picture (and/or other information regarding the first visitor **402a**) with the user group **400**.

[0126] As illustrated in FIG. **10**, the first remote computing device **204a** is not communicatively coupled to the second doorbell **202b**. Embodiments can include receiving, by the first remote computing device **204a**, a second picture **412** taken by the second doorbell **202b** in response to the second doorbell **202b** sending the second picture **412** to a database **390** configured to provide visitor information **406** to the doorbell user group **400**.

[0127] The visitors **402a**, **402b**, **402c** can be the same person. For example, a solicitor knocking on doors of buildings **300a**, **300b**, **300c** in a neighborhood can be detected by motion sensors of many doorbells **202a**, **202b**, **202c**. In some cases, visitors **402b**, **402c** can be different people.

[0128] FIG. **11** illustrates a diagrammatic view of an embodiment that includes a user group **400**. Using a doorbell system **408** can include coupling a first doorbell **202a** having a first camera **208** (labeled in FIG. **1**) to a first building **300a**. The first building **300a** can include a first wireless network **308a** that communicatively couples the first doorbell **202a** to the first remote computing device **204a**, but does not communicatively couple the first doorbell **202a** to a second remote computing device **204b**.

[0129] In some embodiments, the doorbell **202a** is connected to the first wireless **308a**, which connects to the Internet and/or to a cellular network to enable the first doorbell **202a** to be communicatively coupled to the first remote computing device **204a** even when the first remote computing device **204a** is many miles away from the first building **300a**. In some cases, when the first remote computing device **204a** is located within range of the first wireless network **308a** (e.g., inside the first building **300a**), the first wireless network **308a** can directly communicatively couple the first doorbell **202a** to the first remote computing device **204b**. The second building **300b** can include a second wireless network **308b**, which can communicatively couple the second doorbell **202b** to the second remote computing device **204b**. The third building **300c**, to which the third doorbell **202c** can be mechanically coupled, can include a third wireless network **308c**.

[0130] Several embodiments include configuring the first doorbell **202a** to wirelessly communicate with the first remote computing device **204a**. Some embodiments include joining, by the first remote computing device **204a**, a doorbell user group **400** that comprises a second user of the second doorbell **202b** having a second camera coupled to the second building **300b**. The doorbell user group **400** can also include a third user of a third doorbell **202c** having a third camera coupled to the third building **300c**. The second doorbell **202b** can be configured to wirelessly communicate with the second remote computing device **204b**.

[0131] Some embodiments include receiving, by the first remote computing device **204a**, a first visitor picture **410** that was taken by the second doorbell **202b** or the third doorbell **202c**. For example, arrow **416** shows how the first visitor picture **410** from the second doorbell **202b** can be sent to the second remote computing device **204b** and/or to the user group **400** (in some cases without passing through the second remote computing device **204b**). Arrow **418** represents that the second remote computing device **204b** can send the first picture **410** to the user group **400** and/or can send permission for members of the user group **400** to view the first picture **410** to the user group **400**. Arrow **420** represents how the user group **400** can send the first picture **410** to the first remote computing device **204a** and/or to the third remote computing device **204c**.

[0132] As used herein, "user groups" often do not include humans, but instead include devices owned by various humans. For example, a neighborhood association may decide to purchase doorbells **202a**, **202b**, **202c**. Creating a doorbell user group can include communicatively coupling

the doorbells **202a**, **202b**, **202c** and the remote computing devices **204a**, **204b**, **204c** as shown in FIG. **10**. As a result of this communicatively coupling, members of the neighborhood association can share visitor information. A doorbell user group can include the doorbells **202a**, **202b**, **202c** and/or the remote computing devices **204a**, **204b**, **204c**. User groups **400** can include databases **390**, servers **404**, and information **406** regarding the owners of the doorbells **202a**, **202b**, **202c** and the remote computing devices **204a**, **204b**, **204c**. User groups **400** can include the locations of each doorbell in the user group **400**. User groups **400** can store visitor pictures and visitor information for future reference by members of the user groups **400**.

[0133] Referring now to FIG. **11**, embodiments can include receiving from the doorbell user group **400**, by the first remote computing device **204a**, the first visitor picture **410** taken by the second doorbell **202b** even though the first remote computing device **204a** is not communicatively coupled to the second doorbell **202b**. Embodiments can also include receiving, by the first remote computing device **204a**, a first trait regarding a first visitor **402a** in response to a second user of the second remote computing device **204b** inputting the first trait into the second remote computing device **204b**. For example, the second user can label the first visitor **402** as “suspicious” and/or can record a description of the first visitor **402a**.

[0134] Several embodiments include taking a second visitor picture **412**, by the first doorbell **202a**, and sending the second visitor picture **412** to the doorbell user group **400** such that the second remote computing device **204b** and the third remote computing device **204c** can display the second visitor picture **412** even though the second remote computing device **204b** and the third remote computing device **204c** are not communicatively coupled to the first doorbell **202a**.

[0135] The doorbell user group **400** can comprise a database **390** that includes many visitor pictures taken by more doorbells that are communicatively coupled to the doorbell user group **400**. The database **390** can include a first visitor picture **410** taken by the second doorbell **202b** of the second building **300b**. The first doorbell **202a** of the first building **300a** can take a second visitor picture **412**. At least a portion of the doorbell system **408** can determine that the first visitor picture **410** and the second visitor **412** picture show a visitor (i.e., show the same person). Embodiments can include labeling the visitor with solicitor information such that members of the doorbell user group **400** can see the first visitor picture **410** associated with the solicitor information. For example, the remote computing devices **204a**, **204b**, **204c** can download the first visitor picture **410** (or another visitor picture) and information regarding the visitor's solicitation behavior. The remote computing devices **204a**, **204b**, **204c** can display the solicitor information along with the picture of the solicitor.

[0136] Users can type information regarding visitors into their remote computing devices **204a**, **204b**, **204c**. The user group **400** can then share this information with members of the user group **400**. Some embodiments include alerting the members regarding at least a portion of the solicitor information. This alert **422** can include a picture of the solicitor and other information regarding the solicitor. Several embodiments include receiving a solicitor alert **422** in response to the second doorbell **202b** of the second building **300b** and the third doorbell **202c** of the third building **300c** detecting a visitor (e.g., detecting the same person within a predetermined time, which can be within 24 hours).

[0137] The alerts **422** shown in FIG. **11** can also be burglary alerts. Some embodiments include sending an alert **422** to members of the doorbell user group **400** in response to detecting a burglary of the first building **300a**. A remote burglar detection system (e.g., a remote sensor **242** of an alarm system), can detect the burglary. The remote sensor **424** can also be a fire alarm or smoke alarm. Some embodiments include sending an alert **422** to members of the doorbell user group **400** in response to a remote sensor **424** detecting fire and/or smoke.

[0138] A doorbell system **408** can comprise a first doorbell **202a** having a first camera coupled to a first building **300a**, wherein the first doorbell **202a** is communicatively coupled to a first remote computing device **204a**; a second doorbell **202b** having a second camera coupled to a second

building **300b**, wherein the second doorbell **202b** is communicatively coupled to a second remote computing device **204b**; and a doorbell user group **400** comprising a database **390** having images **410**, **412** taken by the first camera and the second camera. The doorbell user group **400** can be communicatively coupled to the first remote computing device **204a** and the second remote computing device **204b**.

[0139] In several embodiments, the first doorbell **202a** is not communicatively coupled to the second remote computing device **204b**, and the second doorbell **202b** is not communicatively coupled to the first remote computing device **204a** while the system **408** is configured such that the first remote computing device **204a** receives a first visitor picture **410** taken by the second doorbell **202b** via the doorbell user group **400**. A second visitor picture **412** can be taken by the first doorbell **202a** and displayed by the second remote computing device **204b**.

[0140] The doorbell system **408** can also include a remote sensor **424** configured to monitor the first building **300a** and detect an unauthorized building intrusion. The doorbell system **408** can include at least one alert **422** sent to the doorbell user group **400** in response to the remote sensor **424** detecting the unauthorized building intrusion.

[0141] Several embodiments include a solicitor alert sent to the doorbell user group **400** in response to a visitor being detected by the first doorbell **202a** at the first building **300a** and by the second doorbell **202b** at the second building **300b**.

[0142] Information collected from the doorbells **202a**, **202b**, **202c** can be analyzed (e.g., by the system). An alert **422** can be sent to remote computing devices **204a**, **204c** in response to analytics based on the collected information.

[0143] Aggregated doorbell information can be analyzed to look for trends and abnormal behaviors that might be correlated with crime or other unwanted behavior. Analytics can include the number of visits or visitors to a particular building, group of buildings, or area. For example, a high number of unique visitors could suggest an illegal business is being operated out of a home (e.g., drug dealing). Analytics can also evaluate how long visitors wait at a door before leaving. In some cases, analytics include determining whether a person with a criminal history or arrest warrant is contacting more than one building in an area within a predetermined time.

[0144] Several embodiments include taking, by the second doorbell **202b**, a first visitor picture **410**; sending, by the second doorbell **202b**, the first visitor picture to the user group; and comparing, by the user group **400**, the first visitor picture **410** to criminal pictures of a database **390**. A moderator (e.g., a person who acts as an administrator of the user group **400**) can compare visitor pictures to determine if the pictures show the same person (e.g., a solicitor) or show a criminal. Some embodiments use computer image recognition to compare images.

[0145] The first building **300a** can include an alarm **434** that can be located remotely relative to the first doorbell **202a**. In some embodiments, the doorbell **202a** is communicatively coupled to the alarm **434**. The alarm **434** can include a speaker and electronics configured to enable the speaker to emit an alarm sound.

[0146] A button on the user interface of the first remote computing device **204a** can enable a user to active the alarm **434**. In response to a signal from the remote sensor **424**, the system **408** can enter an Alarm State, which can include recording, by the first doorbell **202a**, a video; blinking a light **216**, **220** (shown in FIG. 1) of the first doorbell **202a**; and emitting an alarm sound from a speaker **488** (shown in FIG. 4) of the first doorbell **202a**. Other members of the user group **400** can receive an alert **422** regarding the Alarm State. Other members can also see a flashing light and hear the alarm sound from the first doorbell **202a**. As a result, the first doorbell **202a** can serve as a beacon to first responders, which can include neighbors, medical personnel, and law enforcement officers.

[0147] Each member of the user group **400** can choose which members receive alerts, visitor pictures, and visitor information from the choosing member. This approach enables sub-groups within a larger user group.

Interpretation

[0148] None of the steps described herein is essential or indispensable. Any of the steps can be adjusted or modified. Other or additional steps can be used. Any portion of any of the steps, processes, structures, and/or devices disclosed or illustrated in one embodiment, flowchart, or example in this specification can be combined or used with or instead of any other portion of any of the steps, processes, structures, and/or devices disclosed or illustrated in a different embodiment, flowchart, or example. The embodiments and examples provided herein are not intended to be discrete and separate from each other.

[0149] The section headings and subheadings provided herein are nonlimiting. The section headings and subheadings do not represent or limit the full scope of the embodiments described in the sections to which the headings and subheadings pertain. For example, a section titled “Topic 1” may include embodiments that do not pertain to Topic 1 and embodiments described in other sections may apply to and be combined with embodiments described within the “Topic 1” section.

[0150] Some of the devices, systems, embodiments, and processes use computers. Each of the routines, processes, methods, and algorithms described in the preceding sections may be embodied in, and fully or partially automated by, code modules executed by one or more computers, computer processors, or machines configured to execute computer instructions. The code modules may be stored on any type of non-transitory computer-readable storage medium or tangible computer storage device, such as hard drives, solid state memory, flash memory, optical disc, and/or the like. The processes and algorithms may be implemented partially or wholly in application-specific circuitry. The results of the disclosed processes and process steps may be stored, persistently or otherwise, in any type of non-transitory computer storage such as, e.g., volatile or non-volatile storage.

[0151] The various features and processes described above may be used independently of one another, or may be combined in various ways. All possible combinations and subcombinations are intended to fall within the scope of this disclosure. In addition, certain method, event, state, or process blocks may be omitted in some implementations. The methods, steps, and processes described herein are also not limited to any particular sequence, and the blocks, steps, or states relating thereto can be performed in other sequences that are appropriate. For example, described tasks or events may be performed in an order other than the order specifically disclosed. Multiple steps may be combined in a single block or state. The example tasks or events may be performed in serial, in parallel, or in some other manner. Tasks or events may be added to or removed from the disclosed example embodiments. The example systems and components described herein may be configured differently than described. For example, elements may be added to, removed from, or rearranged compared to the disclosed example embodiments.

[0152] Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to

each be present.

[0153] The term “and/or” means that “and” applies to some embodiments and “or” applies to some embodiments. Thus, A, B, and/or C can be replaced with A, B, and C written in one sentence and A, B, or C written in another sentence. A, B, and/or C means that some embodiments can include A and B, some embodiments can include A and C, some embodiments can include B and C, some embodiments can only include A, some embodiments can include only B, some embodiments can include only C, and some embodiments can include A, B, and C. The term “and/or” is used to avoid unnecessary redundancy.

[0154] While certain example embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions disclosed herein. Thus, nothing in the foregoing description is intended to imply that any particular feature, characteristic, step, module, or block is necessary or indispensable. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions, and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions disclosed herein.

Claims

1. A doorbell system, comprising: a first doorbell having a first housing, a first microphone, a first speaker, a first camera, a first transmitter, and a first button, the first doorbell coupled to a first building and communicatively coupled to a first remote computing device; a second doorbell having a second housing, a second microphone, a second speaker, a second camera, a second transmitter, and a second button, the second doorbell coupled to a second building and communicatively coupled to a second remote computing device; a doorbell user group comprising a database having images taken by the first camera and the second camera, wherein the doorbell user group is communicatively coupled to the first remote computing device and the second remote computing device; and a display, on at least one of the first remote computing device and the second remote computing device, configured to show a map that indicates a location of the first doorbell and a location of the second doorbell, wherein the display comprises an activity list comprising an event shared with the doorbell user group.
2. The doorbell system of claim 1, wherein the event is shared with at least one member of the doorbell user group by a user selected from a user of the first doorbell, a user of the second doorbell, and combinations thereof.
3. The doorbell system of claim 2, wherein when the event is shared with the doorbell user group, the doorbell user group is configured to receive data about the event.
4. The doorbell system of claim 3, wherein the data about the event comprises information selected from the group consisting of a photo, a written description, a recorded description, a video, and combinations thereof.
5. The doorbell system of claim 4, wherein when the data about the event indicates suspicious activity, the display is configured to show a tag labeling the event as suspicious.
6. The doorbell system of claim 5, wherein the event can be shared with an entity outside of the doorbell user group.
7. The doorbell system of claim 6, wherein the entity comprises law enforcement.
8. The doorbell system of claim 1, wherein the event comprises detection of a visitor by at least one of the first doorbell and the second doorbell.
9. The doorbell system of claim 2, wherein the event is shared with at least one member of the doorbell user group via email, text message, push notification, or combinations thereof.
10. The doorbell system of claim 1, wherein the event comprises a plurality of events.
11. The doorbell system of claim 10, wherein the map comprises a plurality of icons representing the plurality of events.

12. A method of using a doorbell system, wherein the doorbell system comprises a first remote computing device communicatively coupled to a first doorbell and a second remote computing device communicatively coupled to a second doorbell, the method comprising: capturing, via a first camera of the first doorbell, a first image; sending, via a first transmitter of the first doorbell, the first image to the first remote computing device; capturing, via a second camera of the second doorbell, a second image; sending, via a second transmitter of the second doorbell, the second image to the second remote computing device; creating a doorbell user group comprising the first remote computing device and the second remote computing device; sharing, via the first remote computing device and the second remote computing device, the first image and the second image with the doorbell user group; and displaying, on at least one of the first remote computing device and the second remote computing device, a map indicating a location of the first doorbell and a location of the second doorbell.

13. The method of claim 12, further comprising displaying the first image and the second image on the map, wherein the first image is indicated at the location of the first doorbell and the second image is indicated at the location of the second doorbell.

14. The method of claim 12, further comprising displaying, via the map, an activity list comprising an event.

15. The method of claim 14, wherein the event comprises detection of a visitor by at least one of the first doorbell and the second doorbell, and wherein the event is represented by an icon displayed on the map.

16. The method of claim 15, further comprising sharing the event with the doorbell user group upon one of the first remote computing device and the second remote computing device receiving a user selection of the icon.

17. The method of claim 16, wherein sharing the event with the doorbell user group comprises sharing data about the event, wherein the data about the event comprises information selected from the group consisting of a photo, a written description, a recorded description, a video, and combinations thereof.

18. The method of claim 17, further comprising determining that the event indicates suspicious activity and displaying, on the map, a tag labeling the event as suspicious.

19. The method of claim 18, further comprising reporting, via one of the first remote computing device and the second remote computing device, the event to law enforcement.

20. The method of claim 18, further comprising sending, via one of the first remote computing device and the second remote computing device, a distress signal to the doorbell user group, wherein the distress signal comprises a push notification received by at least one of the first remote computing device and the second remote computing device.
